FormFix AI

Executive Summary & Technology Flow Guide

What is FormFix AI? FormFix AI is an AI-powered document action assistant. When students and citizens receive complicated notices (scholarships, college admissions, circulars, job descriptions), FormFix AI does not just summarize the text—it converts the document into **Deadlines**, **Required Documents**, and an **Interactive Action Plan** that users can check off as they complete each task.

1. Technology Stack & Why Each Tool is Used

Technology	What It Does	Why We Used It (Advantage)
Streamlit (Python UI)	Powers the web application, file uploader, progress bar, interactive checkboxes, and chat box.	Allows building interactive, beautiful web apps in 100% Python without complex React or Node.js setup.
PyMuPDF (fitz)	Reads and extracts digital text from uploaded PDF documents with exact typography.	Extremely fast native C engine (20x faster than standard PDF libraries); avoids lag during document uploads.
Pillow & Tesseract (OCR Engine)	Enhances image contrast and reads text from scanned documents, photocopies, or phone images.	Fallback safety net: if a student uploads a scanned photo or circular image, OCR extracts the text automatically.
Groq Cloud API (Llama 3 / Qwen)	Understands the document text, extracts structured JSON data, and powers the Q&A; chat assistant.	Provides ultra-fast AI inference (over 300 tokens/sec), generating results in 1-2 seconds with zero hallucinations.
Session State (Streamlit Storage)	Saves completed checklist checkboxes and chat history while using the app.	Keeps user documents strictly in memory without needing a separate database (100% private by design).

2. How It Works (Step-by-Step Flow)

Step 01	**Upload Document** User uploads an official notice or form (PDF, PNG, or JPG) into the web interface.
Step 02	**Smart Text Extraction** PyMuPDF reads the PDF text. If the document is scanned or an image, OCR triggers automatically.
Step 03	**Text Cleaning** Whitespace and broken lines are cleaned using regex to keep the text compact and clear.
Step 04	**AI Structured Extraction (Groq)** The clean text is sent to the LLM. It extracts dates, documents, confusing terms, and action steps into pure JSON.
Step 05	**Action Dashboard** User receives an interactive checklist with checkboxes, prominent deadline banner, and required documents.
Step 06	**Ask FormFix AI** User can ask questions (e.g. 'What is the deadline?'). The AI answers strictly from the document without making up facts.

3. The 5 Core Results Generated for the User

1. **Critical Deadline Banner:** High-visibility alert displaying the exact cutoff date and time (e.g. 15 October 2026, 5:00 PM IST) so students never miss submissions.
2. **Interactive Action Plan (What to Do Next):** Practical checklist with real-time percentage progress (e.g. 3 of 7 tasks done - 42%). Users tick off items as they collect documents.
3. **Required Supporting Documents:** Itemized list of mandatory certificates (Marksheets, Bonafide Certificate, Income Certificate, Fee Receipts).

4. **Confusing Fields Decoded:** Explains institutional jargon in simple everyday words and matches each with the required certificate.

5. **Zero-Hallucination Grounded Chat:** Interactive conversational assistant that only answers from the document. If information is missing, it explicitly states so.

4. Quick Hackathon Pitch (Talking Points)

- **Not Just a Summarizer:** Summarizers give paragraphs of text. FormFix AI gives an actionable checklist.
- **Speed:** Powered by Groq, full document processing takes under 2 seconds.
- **Accuracy & Trust:** Strict anti-hallucination prompting ensures no made-up deadlines or documents.
- **Zero Deployment Friction:** Built with pure Python & Streamlit, deployable in 1 click to Streamlit Cloud.